

Fuzzy Description Logics and the Semantic Web

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“Calla is a **very large**, **long** white flower on **thick** stalks”

Outline

1 The Semantic Web and Ontologies

- The Semantic Web Vision
- Ontologies

2 Description Logics

- DLs Basics

3 Fuzzy Description Logics

- A clarification: Uncertainty v.s. Imprecision
- Examples of applications
- Top- k retrieval in DLs
- Propositional Fuzzy Logics Basics
- Predicate Fuzzy Logics Basics
- Fuzzy DLs Basics
- Towards fuzzy OWL Lite and OWL DL

The Semantic Web and Ontologies (excerpt)

The Semantic Web Vision

- The WWW as we know it now
 - ▶ **1st generation** web mostly handwritten HTML pages
 - ▶ **2nd generation** (current) web often machine generated/active
 - ▶ Both intended for direct human processing/interaction
- In **next generation** web, **resources** should be more accessible to automated processes
 - ▶ To be achieved via **semantic markup**
 - ▶ **Metadata** annotations that describe content/function

Ontologies

- Semantic markup must be **meaningful** to automated processes
- Ontologies will play a key role
 - ▶ Source of **precisely defined** terms (vocabulary)
 - ▶ Can be **shared** across applications (and humans)
- Ontology typically consists of:
 - ▶ **Hierarchical** description of important **concepts** in domain
 - ▶ Descriptions of **properties** of instances of each concept
- Ontologies can be used, e.g.
 - ▶ To facilitate agent-agent communication in **e-commerce**
 - ▶ In semantic based **search**
 - ▶ To provide richer **service descriptions** that can be more flexibly interpreted by intelligent agents

Example Ontology

- Vocabulary and meaning (“definitions”)
 - ▶ **Elephant** is a concept whose members are a kind of animal
 - ▶ **Herbivore** is a concept whose members are exactly those animals who eat only plants or parts of plants
 - ▶ **Adult_Elephant** is a concept whose members are exactly those elephants whose age is greater than 20 years
- Background knowledge/constraints on the domain (“general axioms”)
 - ▶ **Adult_Elephants** weigh at least 2,000 kg
 - ▶ All **Elephants** are either **African_Elephants** or **Indian_Elephants**
 - ▶ No individual can be both a **Herbivore** and a **Carnivore**

Example Ontology (Protégé)

The screenshot displays the Protégé 3.0 ontology editor interface. The title bar indicates the file path: `(file:/Users/horrocks/Software/OilEd/ontologies/elephants.pprj, OWL Files (.owl or .rdf))`. The menu bar includes **File**, **Edit**, **Project**, **OWL**, **Wizards**, **Code**, **Window**, and **Help**. The toolbar contains various icons for file operations and ontology editing.

The main workspace is divided into several panes:

- Left Pane (Subclass Relationship):** Shows the asserted hierarchy for the project 'elephants'. The hierarchy starts with `owl:Thing` and branches into `ns0:animal`, which further branches into `ns0:african_animal`, `ns0:asian_animal`, `ns0:carnivore`, and `ns0:elephant`. The `ns0:elephant` class is expanded, showing its subclasses: `ns0:adult_elephant`, `ns0:african_elephant`, `ns0:indian_elephant`, `ns0:kenyan_elephant`, `ns0:giraffe` (selected), `ns0:herbivore`, `ns0:large_animal`, `ns0:lion`, `ns0:branch`, and `ns0:continent`.
- Top Center Pane (Class Editor):** Displays the editor for the selected class `ns0:giraffe` (instance of `owl:Class`). It includes tabs for **Name**, **SameAs**, and **DifferentFrom**. The **Name** tab shows the class name `ns0:giraffe` and an `rdfs:comment` of "Funny looking things with long necks". The **Annotations** tab shows a table with the following data:

Property	Value	La...
<code>rdfs:comment</code>	"Funny looking...	
- Bottom Center Pane (Asserted Conditions):** Shows the asserted conditions for the class. It includes a section for **NECESSARY & SUFFICIENT** conditions and a section for **NECESSARY** conditions. The necessary conditions listed are `ns0:animal` and `ns0:eats ns0:leaf`.
- Right Pane (Disjoints):** Shows the disjoints for the class. It includes a section for **Disjoints** and a section for **Logic View** and **Properties View**.

Ontology Description Languages

- Should be **sufficiently expressive** to capture most useful aspects of domain knowledge representation
- Reasoning in it should be **decidable** and **efficient**
- Many different languages has been proposed: RDF, RDFS, OIL, DAML+OIL
- OWL (**O**ntology **W**eb **L**anguage) is the current emerging language. There are three species of OWL
 - ▶ OWL full is union of OWL syntax and RDF (but, undecidable)
 - ▶ OWL DL restricted to FOL fragment (reasoning problem in NEXPTIME)
 - ★ based on **SHIQ Description Logic** ($\mathcal{ALCHIQ}_+$)
 - ▶ OWL Lite is “easier to implement” subset of OWL DL (reasoning problem in EXPTIME)
 - ★ based on **SHIF Description Logic** ($\mathcal{ALCHIF}_+$)
- SWRL, a **S**emantic **W**eb **R**ule **L**anguage combines OWL and RuleML (not addressed here)

Description Logics (excerpt)

Description Logics Basics

(the logics behind OWL, <http://dl.kr.org/>)

- **Concept/Class**: names are equivalent to unary predicates
 - ▶ In general, concepts equiv to formulae with one free variable
- **Role or attribute**: names are equivalent to binary predicates
 - ▶ In general, roles equiv to formulae with two free variables
- **Taxonomy**: Concept and role hierarchies can be expressed
- **Individual**: names are equivalent to constants
- **Operators**: restricted so that:
 - ▶ Language is decidable and, if possible, of low complexity
 - ▶ No need for explicit use of variables
 - ★ Restricted form of $\exists$ and $\forall$
 - ▶ Features such as counting can be succinctly expressed

The DL Family

- A given DL is defined by set of concept and role forming operators
- Basic language: $\mathcal{ALC}$ ($\mathcal{A}$ ttributive $\mathcal{L}$ anguage with $\mathcal{C}$ omplement)

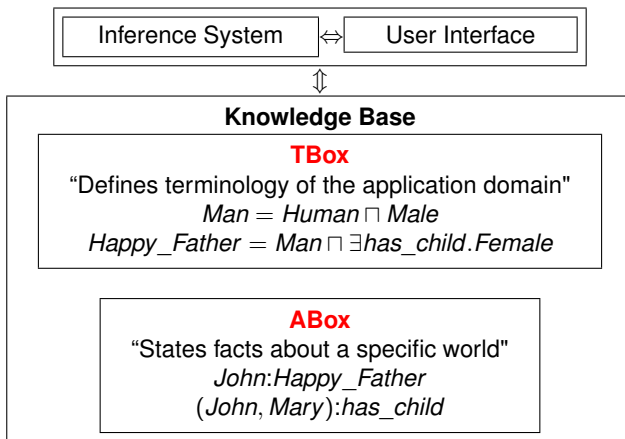
Syntax	Semantics	Example
$C, D \rightarrow$	$\top$	$\top(x)$
	$\perp$	$\perp(x)$
	A	$A(x)$
$C \sqcap D$	$C(x) \wedge D(x)$	<i>Human</i>
$C \sqcup D$	$C(x) \vee D(x)$	<i>Human</i> $\sqcap$ <i>Male</i>
$\neg C$	$\neg C(x)$	<i>Nice</i> $\sqcap$ <i>Rich</i>
$\exists R.C$	$\exists y. R(x, y) \wedge C(y)$	$\neg$ <i>Meat</i>
$\forall R.C$	$\forall y. R(x, y) \Rightarrow C(y)$	$\exists$ <i>has_child.Blond</i>
$C \sqsubseteq D$	$\forall x. C(x) \Rightarrow D(x)$	$\forall$ <i>has_child.Human</i>
$a:C$	$C(a)$	<i>Happy_Father</i> $\sqsubseteq$ <i>Man</i> $\sqcap$ $\exists$ <i>has_child.Female</i>
		<i>John:Happy_Father</i>

DLs Semantics

- **Interpretation**: $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$, where $\Delta^{\mathcal{I}}$ is the **domain** (a non-empty set), $\cdot^{\mathcal{I}}$ is an **interpretation function** that maps:
 - ▶ **Concept** (class) name A into a function $A^{\mathcal{I}}: \Delta^{\mathcal{I}} \rightarrow \{0, 1\}$
 - ▶ **Role** (property) name R into a function $R^{\mathcal{I}}: \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}} \rightarrow \{0, 1\}$
 - ▶ **Individual** name a into an element of $\Delta^{\mathcal{I}}$
- $\mathcal{ALC}$ mapping to FOL:

$\top(x)$	$\mapsto$	1	$\perp(x)$	$\mapsto$	0
$A(x)$	$\mapsto$	$A(x)$	$(C_1 \sqcap C_2)(x)$	$\mapsto$	$C_1(x) \wedge C_2(x)$
$(C_1 \sqcup C_2)(x)$	$\mapsto$	$C_1(x) \vee C_2(x)$	$(\neg C)(x)$	$\mapsto$	$\neg C(x)$
$(\exists R.C)(x)$	$\mapsto$	$\exists y. R(x, y) \wedge C(y)$	$(\forall R.C)(x)$	$\mapsto$	$\forall y. R(x, y) \Rightarrow C(y)$
$C \sqsubseteq D$	$\mapsto$	$\forall x. C(x) \Rightarrow D(x)$	$a:C$	$\mapsto$	$C(a)$

Description Logic System



Note on DL naming

$\mathcal{AL}$: $C, D \rightarrow \top \mid \perp \mid A \mid C \sqcap D \mid \neg A \mid \exists R. \top \mid \forall R. C$

$\mathcal{C}$: Concept negation, $\neg C$. Thus, $\mathcal{ALC} = \mathcal{AL} + \mathcal{C}$

$\mathcal{S}$: Used for $\mathcal{ALC}$ with transitive roles $\mathcal{R}_+$

$\mathcal{U}$: Concept disjunction, $C_1 \sqcup C_2$

$\mathcal{E}$: Existential quantification, $\exists R. C$

$\mathcal{H}$: Role inclusion axioms, $R_1 \sqsubseteq R_2$, e.g. *is_component_of* $\sqsubseteq$ *is_part_of*

$\mathcal{N}$: Number restrictions, $(\geq n R)$ and $(\leq n R)$, e.g. $(\geq 3 \text{ has_Child})$ (has at least 3 children)

$\mathcal{Q}$: Qualified number restrictions, $(\geq n R. C)$ and $(\leq n R. C)$,
e.g. $(\leq 2 \text{ has_Child. Adult})$ (has at most 2 adult children)

$\mathcal{O}$: Nominals (singleton class), $\{a\}$, e.g. $\exists \text{has_child.}\{mary\}$.

Note: $a:C$ equiv to $\{a\} \sqsubseteq C$ and $(a, b):R$ equiv to $\{a\} \sqsubseteq \exists R. \{b\}$

$\mathcal{I}$: Inverse role, R^- , e.g. *isPartOf* = *hasPart*⁻

$\mathcal{F}$: Functional role, f , e.g. *functional*(*hasAge*)

$\mathcal{R}_+$: transitive role, e.g. *transitive*(*isPartOf*)

For instance,

$$\begin{aligned} SHIF &= S + \mathcal{H} + \mathcal{I} + \mathcal{F} = \mathcal{ALCR}_+HIF \\ SHOIN &= S + \mathcal{H} + \mathcal{O} + \mathcal{I} + \mathcal{N} = \mathcal{ALCR}_+HOIN \end{aligned}$$

OWL-Lite (EXPTIME)

OWL-DL (NEXPTIME)



Excerpt of **pizza** ontology ... (according to University of Manchester)

PizzaFruttiDiMare = *Pizza*
 $\sqcap \exists \text{hasTopping} . \text{MixedSeafoodTopping}$
 $\sqcap \exists \text{hasTopping} . \text{GarlicTopping}$
 $\sqcap \exists \text{hasTopping} . \text{TomatoTopping}$
 $\sqcap \forall \text{hasTopping} . (\text{MixedSeafoodTopping} \sqcup \text{GarlicTopping} \sqcup \text{TomatoTopping})$
 $\sqcap \exists \text{hasBase} . \text{PizzaBase}$

PizzaBase $\sqsupseteq$ *DeepPanBase* $\sqcup$ *ThinAndCrispyBase*

MixedSeafoodTopping $\sqsubseteq$ *FishTopping*

FishTopping $\sqsubseteq$ *PizzaTopping* $\sqcap \exists \text{hasSpiciness} . \text{Mild}$

disjoint(*FishTopping*, *MeatTopping*, *HerbSpiceTopping*)

functional(*hasSpiciness*)

Topping $\sqsubseteq$ $\forall \text{hasSpiciness} . (\text{Hot} \sqcup \text{Medium} \sqcup \text{Mild})$

Concrete domains

- **Concrete domains**: integers, strings, ...
- Clean separation between “object” classes and concrete domains
 - ▶ $D = \langle \Delta_D, \Phi_D \rangle$
 - ▶ Δ_D is an interpretation domain
 - ▶ Φ_D is the set of concrete domain predicates d with a predefined arity n and **fixed** interpretation $d^D: \Delta_D^n \rightarrow \{0, 1\}$
 - ▶ Concrete properties: $R^{\mathcal{I}}: \Delta^{\mathcal{I}} \times \Delta_D \rightarrow \{0, 1\}$

(tim, 14):hasAge

(sf, “SoftComputing”):hasAcronym

(source1, “ComputerScience”):isAbout

(service2, “InformationRetrievalTool”):Matches

- Philosophical reasons: concrete domains structured by **built-in predicates**
- Practical reasons:
 - ▶ language remains **simple and compact**
 - ▶ **Semantic integrity** of language not compromised
 - ▶ **Implementability** not compromised – can use hybrid reasoner
 - ★ Only need sound and complete decision procedure for $d_1^{\mathcal{I}} \wedge \dots \wedge d_n^{\mathcal{I}}$, where d_i is a (possibly negated) concrete property
- Notation: (D) . E.g., $\mathcal{ALC}(D)$ is $\mathcal{ALC}$ + concrete domains



Abstract Syntax	DL Syntax	Example
Descriptions (C)		
A (URI reference) owl:Thing owl:Nothing	A $\top$ $\perp$	Conference
intersectionOf($C_1 C_2 \dots$) unionOf($C_1 C_2 \dots$) complementOf(C) oneOf($a_1 \dots$)	$C_1 \sqcap C_2$ $C_1 \sqcup C_2$ $\neg C$ $\{a_1, \dots\}$	Reference $\sqcap$ Journal Organization $\sqcup$ Institution $\neg$ MasterThesis $\{\text{"WISE"}, \text{"ISWC"}, \dots\}$
restriction(R someValuesFrom(C)) restriction(R allValuesFrom(C)) restriction(R hasValue(a)) restriction(R minCardinality(n)) restriction(R maxCardinality(n))	$\exists R.C$ $\forall R.C$ $R : a$ $(\geq n R)$ $(\leq n R)$	$\exists$ parts.InCollection $\forall$ date.Date date : 2005 ≥ 1 location ≤ 1 publisher
restriction(U someValuesFrom(D)) restriction(U allValuesFrom(D)) restriction(U hasValue(v)) restriction(U minCardinality(n)) restriction(U maxCardinality(n))	$\exists U.D$ $\forall U.D$ $U : v$ $(\geq n U)$ $(\leq n U)$	$\exists$ issue.integer $\forall$ name.string series : "LNCS" ≥ 1 title ≤ 1 author

Abstract Syntax	DL Syntax	Example
Axioms		
Class(<i>A</i> partial $C_1 \dots C_n$)	$A \sqsubseteq C_1 \sqcap \dots \sqcap C_n$	$Human \sqsubseteq Animal \sqcap Biped$
Class(<i>A</i> complete $C_1 \dots C_n$)	$A = C_1 \sqcap \dots \sqcap C_n$	$Man = Human \sqcap Male$
EnumeratedClass(<i>A</i> $o_1 \dots o_n$)	$A = \{o_1\} \sqcup \dots \sqcup \{o_n\}$	$RGB = \{r\} \sqcup \{g\} \sqcup \{b\}$
SubClassOf($C_1 C_2$)	$C_1 \sqsubseteq C_2$	
EquivalentClasses($C_1 \dots C_n$)	$C_1 = \dots = C_n$	
DisjointClasses($C_1 \dots C_n$)	$C_i \sqcap C_j = \perp, i \neq j$	$Male \sqsubseteq \neg Female$
ObjectProperty(<i>R</i> super (R_1)... super (R_n))	$R \sqsubseteq R_i$	$HasDaughter \sqsubseteq hasChild$
domain(C_1)...domain(C_n)	$(\geq 1 R) \sqsubseteq C_i$	$(\geq 1 hasChild) \sqsubseteq Human$
range(C_1)...range(C_n)	$\top \sqsubseteq \forall R.D_i$	$\top \sqsubseteq \forall hasChild.Human$
[inverseof(R_0)]	$R = R_0^-$	$hasChild = hasParent^-$
[symmetric]	$R = R^-$	$similar = similar^-$
[functional]	$\top \sqsubseteq (\leq 1 R)$	$\top \sqsubseteq (\leq 1 hasMother)$
[Inversefunctional]	$\top \sqsubseteq (\leq 1 R^-)$	
[Transitive]	$Tr(R)$	$Tr(ancestor)$
SubPropertyOf($R_1 R_2$)	$R_1 \sqsubseteq R_2$	
EquivalentProperties($R_1 \dots R_n$)	$R_1 = \dots = R_n$	$cost = price$
AnnotationProperty(<i>S</i>)		

Abstract Syntax	DL Syntax	Example
DatatypeProperty(U super (U_1)... super (U_n)) domain(C_1)...domain(C_n) range(D_1)...range(D_n) [functional] SubPropertyOf($U_1 U_2$) EquivalentProperties(U_1 ... U_n)	$U \sqsubseteq U_i$ $(\geq 1 U) \sqsubseteq C_i$ $\top \sqsubseteq \forall U. D_i$ $\top \sqsubseteq (\leq 1 U)$ $U_1 \sqsubseteq U_2$ $U_1 = \dots = U_n$	$(\geq 1 \text{ hasAge}) \sqsubseteq \text{Human}$ $\top \sqsubseteq \forall \text{ hasAge. posInteger}$ $\top \sqsubseteq (\leq 1 \text{ hasAge})$ $\text{hasName} \sqsubseteq \text{hasFirstName}$
Individuals		
Individual(o type (C_1)... type (C_n)) value($R_1 o_1$)...value($R_n o_n$) value($U_1 v_1$)...value($U_n v_n$) SameIndividual(o_1 ... o_n) DifferentIndividuals(o_1 ... o_n)	$o:C_i$ $(o, o_i):R_i$ $(o, v_i):U_i$ $o_1 = \dots = o_n$ $o_i \neq o_j, i \neq j$	tim:Human $(\text{tim}, \text{mary}):\text{hasChild}$ $(\text{tim}, 14):\text{hasAge}$ $\text{president_Bush} = \text{G.W.Bush}$ $\text{john} \neq \text{peter}$

XML representation of OWL statements

E.g., $Person \sqcap \forall hasChild.(Doctor \sqcup \exists hasChild.Doctor)$:

```
<owl:Class>
  <owl:intersectionOf rdf:parseType=" collection">
    <owl:Class rdf:about="#Person"/>
    <owl:Restriction>
      <owl:onProperty rdf:resource="#hasChild"/>
      <owl:allValuesFrom>
        <owl:unionOf rdf:parseType=" collection">
          <owl:Class rdf:about="#Doctor"/>
          <owl:Restriction>
            <owl:onProperty rdf:resource="#hasChild"/>
            <owl:someValuesFrom rdf:resource="#Doctor"/>
          </owl:Restriction>
        </owl:unionOf>
      </owl:allValuesFrom>
    </owl:Restriction>
  </owl:intersectionOf>
</owl:Class>
```

Fuzzy Description Logics

Objective

- To extend classical DLs and LPs towards the representation of and reasoning with **vague concepts**
- To show some applications
- Development of practical reasoning algorithms

A clarification: Uncertainty v.s. Imprecision

- **Uncertainty theory**: statements rather than being either true or false, are true or false to some **probability** or **possibility/necessity**
 - ▶ E.g., “It is possible that it will rain tomorrow”
 - ▶ Usually we have a possible world semantics with a distribution over possible worlds:

$$W = \{I \text{ classical interpretation}\}, \quad I(\varphi) \in \{0, 1\}$$

$$\mu: W \rightarrow [0, 1], \quad \mu(I) \in [0, 1]$$

$$\text{e.g. } Pr(\phi) = \sum_{I \models \phi} \mu(I), \quad Poss(\phi) = \sup_{I \models \phi} \mu(I)$$

- **Imprecision theory**: statements are true to some degree which is taken from a truth space
 - ▶ E.g., “Chinese items are **cheap**”
 - ▶ **Truth space**: set of truth values L and an partial order $\leq$
 - ▶ **Many-valued Interpretation**: a function I mapping formulae into L , i.e. $I(\varphi) \in L$
 - ▶ **Fuzzy Logic**: $L = [0, 1]$
- **Uncertainty and imprecision theory**: “It is **possible** that it will be **hot** tomorrow”
- In this work we deal with **imprecision** and, thus, statements have a degree of truth.

Examples of applications (Ontology mediated data access)

Example (Top-k retrieval)

Hotel $\sqsubseteq \exists \text{hasLoc}$
Conference $\sqsubseteq \exists \text{hasLoc}$
Hotel $\sqsubseteq \neg \text{Conference}$

<i>HotelID</i>	<i>hasLoc</i>	<i>ConferenceID</i>	<i>hasLoc</i>
<i>h1</i>	<i>h1</i>	<i>c1</i>	<i>c1</i>
<i>h2</i>	<i>h1</i>	<i>c2</i>	<i>c1</i>
$\vdots$	$\vdots$	$\vdots$	$\vdots$

<i>hasLoc</i>	<i>hasLoc</i>	<i>distance</i>	<i>hasLoc</i>	<i>hasLoc</i>	<i>close</i>
<i>h1</i>	<i>c1</i>	300	<i>h1</i>	<i>c1</i>	0.7
<i>h1</i>	<i>c2</i>	500	<i>h1</i>	<i>c2</i>	0.5
<i>h2</i>	<i>c1</i>	750	<i>h2</i>	<i>c1</i>	0.25
<i>h2</i>	<i>c2</i>	800	<i>h2</i>	<i>c2</i>	0.2
$\vdots$	$\vdots$		$\vdots$	$\vdots$	

“Find hotels close to the university of Bari”

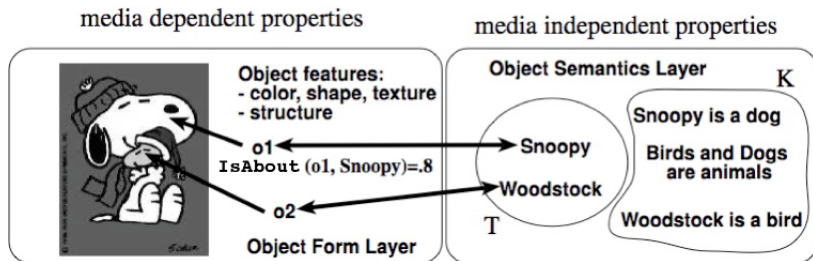
$$q(h) \leftarrow \text{hasLocation}(h, h1) \wedge \text{hasLocation}(\text{uniba}, c1) \wedge \text{close}(h1, c1)$$

Top-k Fuzzy Retrieval: Retrieve the top-k ranked tuples that instantiate the query q w.r.t. the best truth value bound

Note: retrieving all tuples, ranking them and then selecting the top-k ones is not feasible in practice (millions of tuples in the database)



Example (Logic-based information retrieval model, Top-k retrieval)



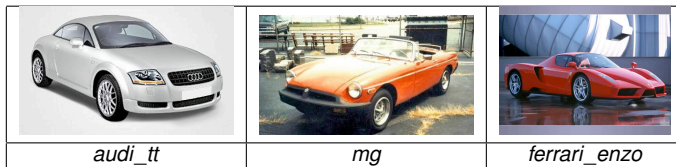
Bird $\sqsubset$ *Animal*
Dog $\sqsubset$ *Animal*
snoopy $:$ *Dog*
woodstock $:$ *Bird*

ImageRegion	Object ID	isAbout
o1	snoopy	0.8
o2	woodstock	0.7
⋮	⋮	

“Find image regions about animals”

$Query(ir) \leftarrow ImageRegion(ir) \wedge isAbout(ir, x) \wedge Animal(x)$

Example (Graded Entailment)

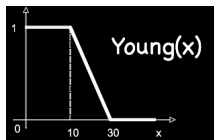


<i>Car</i>	<i>speed</i>
<i>audi_tt</i>	243
<i>mg</i>	≤ 170
<i>ferrari_enzo</i>	≥ 350

SportsCar = *Car* $\sqcap \exists \text{hasSpeed}.\text{very}(\text{High})$

$\mathcal{K} \models \langle \text{ferrari_enzo}:\text{SportsCar}, 1 \rangle$
 $\mathcal{K} \models \langle \text{audi_tt}:\text{SportsCar}, 0.92 \rangle$
 $\mathcal{K} \models \langle \text{audi_tt}:\neg\text{SportsCar}, 0.72 \rangle$

Example (Graded Subsumption)

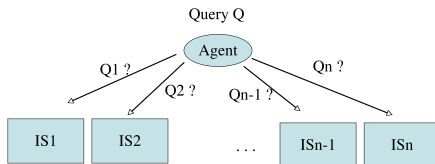


$$\begin{aligned} Minor &= Person \sqcap \exists hasAge. \leq_{18} \\ YoungPerson &= Person \sqcap \exists hasAge. Young \end{aligned}$$

$$\mathcal{K} \models \langle Minor \sqsubseteq YoungPerson, 0.2 \rangle$$

Note: without an explicit membership function of *Young*, this inference cannot be drawn

Example (Distributed Information Retrieval)



Then the agent has to perform **automatically** the following steps:

- 1 the agent has to select a subset of relevant resources $\mathcal{S}' \subseteq \mathcal{S}$, as it is not reasonable to assume to access to and query all resources (**resource selection/resource discovery**);
- 2 for every selected source $\mathcal{S}_i \in \mathcal{S}'$ the agent has to reformulate its information need Q_A into the query language $\mathcal{L}_i$ provided by the resource (**schema mapping/ontology alignment**);
- 3 the results from the selected resources have to be merged together (**data fusion/rank aggregation**)

- **Resource selection/resource discovery:**

- ▶ Use techniques from Distributed Information Retrieval, e.g. CORI

- **Schema mapping/ontology alignment:**

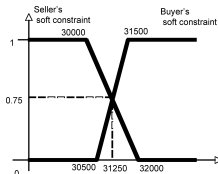
- ▶ Use machine learning techniques, (implemented in oMap)
 - ★ Learns automatically weighted rules, like (aligning Google- Yahoo directories)

$$\text{Mechanical_and_Aerospace_Engineering}(d) \leftarrow 0.81 \cdot \text{Aeronautics_and_Astronautics}(d)$$

- **Data fusion/rank aggregation:**

- ▶ Use techniques from Information Retrieval and/or Voting Systems, e.g. CombMNZ or Borda count

Example (Negotiation)



- a car seller sells an Audi TT for \$31500, as from the catalog price.
- a buyer is looking for a sports-car, but wants to pay not more than around \$30000
- classical DLs: the problem relies on the crisp conditions on price
- more fine grained approach: to consider prices as fuzzy sets (as usual in negotiation)
 - ▶ seller may consider optimal to sell above \$31500, but can go down to \$30500
 - ▶ the buyer prefers to spend less than \$30000, but can go up to \$32000

$$AudiTT = SportsCar \sqcap \exists hasPrice. R(x; 30500, 31500)$$

$$Query = SportsCar \sqcap \exists hasPrice. L(x; 30000, 32000)$$

- ▶ highest degree to which the concept

$$C = AudiTT \sqcap Query$$

is satisfiable is 0.75 (the possibility that the Audi TT and the query **matches** is 0.75)

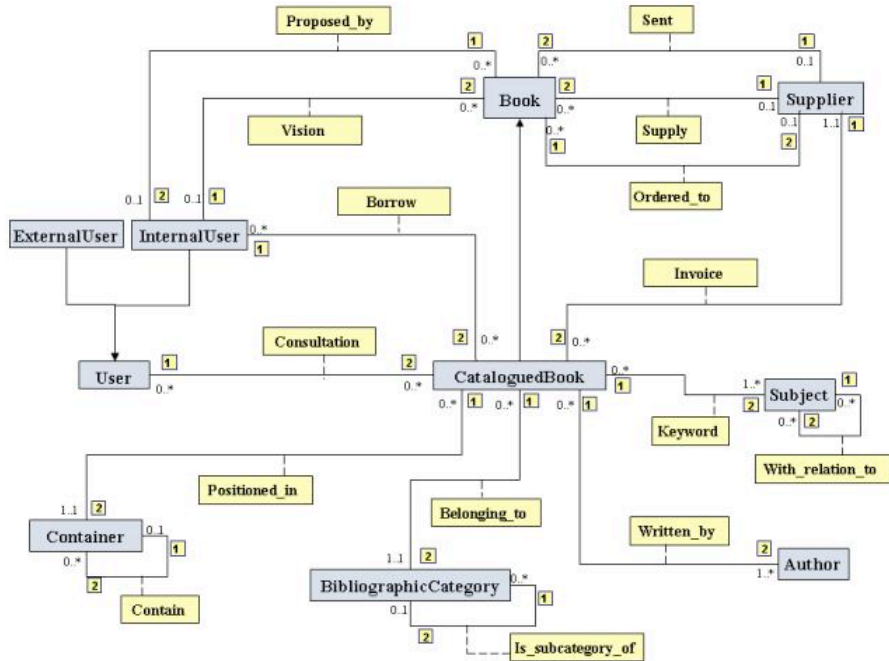
- ▶ the car may be sold at \$31250

Top- k retrieval in DLs: the case of DL-Lite

- **DL-Lite**: a simple, but interesting DL
- Captures important subset of UML/ER diagrams
- Computationally tractable DL to query large databases
- **Sub-linear**, i.e. LOGSpace in data complexity
 - ▶ (same cost as for SQL)
- Good for **very large** database tables, with limited declarative schema design

- **Knowledge base:** $\mathcal{K} = \langle \mathcal{T}, \mathcal{A} \rangle$, where $\mathcal{T}$ and $\mathcal{A}$ are finite sets of axioms and assertions
- **Axiom:** $B \sqsubseteq C$ (inclusion axiom)
 $fun(R)$ (functionality axiom)
- **Note for inclusion axioms:** the language for left hand side is different from the one for right hand side
- **Concepts:**

$$\begin{array}{lcl} B & \rightarrow & A \mid \exists R \\ C & \rightarrow & B \mid \neg B \mid C_1 \sqcap C_2 \\ R & \rightarrow & P \mid P^- \end{array}$$
- **Assertion:** $a:B$ or of the form $(a, b):P$ (assertions are stored in relational tables)
- **Conjunctive query:** $q(\mathbf{x}) \leftarrow \exists \mathbf{y}. conj(\mathbf{x}, \mathbf{y})$
 $conj$ is an **aggregation** of expressions of the form $B(z)$ or $P(z_1, z_2)$,



- Examples:
 - isa* $CatalogueBook \sqsubseteq Book$
 - disjointness* $Book \sqsubseteq \neg Author$
 - constraints* $CatalogueBook \sqsubseteq \exists positioned_In$
 - role – typing* $\exists positioned_In \sqsubseteq Container$
 - functional* $fun(positioned_In)$
 - constraints* $Author \sqsubseteq \exists written_By^-$
 $\exists written_By \sqsubseteq CatalogueBook$
 - assertion* $Romeo_and_Juliet:CatalogueBook$
 $(Romeo_and_Juliet, Shakespeare):written_By$
 - query* $q(x, y) \leftarrow CataloguedBook(x), Ordered_to(x, y)$
- Consistency check is linear time in the size of the KB
- Query answering in linear in in the size of the number of assertions

Top- k retrieval in DL-Lite

- We extend the query formalism:
 - ▶ conjunctive queries, where fuzzy predicates may appear
- **conjunctive query**

$$q(\mathbf{x}, s) \leftarrow \exists \mathbf{y}. \text{conj}(\mathbf{x}, \mathbf{y}) \wedge s = f(p_1(\mathbf{z}_1), \dots, p_n(\mathbf{z}_n))$$

- 1 $\mathbf{x}$ are the *distinguished variables*;
- 2 s is the *score variable*, taking values in $[0, 1]$;
- 3 $\mathbf{y}$ are existentially quantified variables, called *non-distinguished variables*;
- 4 $\text{conj}(\mathbf{x}, \mathbf{y})$ is a conjunction of atoms of the form $A(z)$, or $P(z, z')$, where A and P are respectively an atomic concept and a role (but, not inverse role) in $\mathcal{K}$;
- 5 z, z' are constants in $\mathcal{K}$ or variables in $\mathbf{x}$ or $\mathbf{y}$;
- 6 $\mathbf{z}_i$ are tuples of constants in $\mathcal{K}$ or variables in $\mathbf{x}$ or $\mathbf{y}$;
- 7 p_i is an n_i -ary *fuzzy predicate* assigning to each n_i -ary tuple $\mathbf{c}_i$ the score $p_i(\mathbf{c}_i) \in [0, 1]$;
- 8 f is a monotone *scoring function* $f: [0, 1]^n \rightarrow [0, 1]$, which combines the scores of the n fuzzy predicates $p_i(\mathbf{c}_i)$

Example:

$Hotel \sqsubseteq \exists HasHLoc$
 $Hotel \sqsubseteq \exists HasHPrice$
 $Conference \sqsubseteq \exists HasCLoc$
 $Hotel \sqsubseteq \neg Conference$

HasHLoc		HasCLoc		HasHPrice	
HotelID	HasLoc	ConfID	HasLoc	HotelID	Price
<i>h1</i>	<i>hl1</i>	<i>c1</i>	<i>cl1</i>	<i>h1</i>	150
<i>h2</i>	<i>hl2</i>	<i>c2</i>	<i>cl2</i>	<i>h2</i>	200
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

$$\begin{aligned}
 q(h, s) \leftarrow & \text{HasHLoc}(h, hl) \wedge \text{HasHPrice}(h, p) \wedge \\
 & \text{HasCLoc}(c1, cl) \wedge s = \text{cheap}(p) \cdot \text{close}(hl, cl) .
 \end{aligned}$$

where the fuzzy predicates *cheap* and *close* are defined as

$$\begin{aligned}
 \text{close}(hl, cl) &= \max(0, 1 - \frac{\text{distance}(hl, cl)}{2000}) \\
 \text{cheap}(\text{price}) &= \max(0, 1 - \frac{\text{price}}{300})
 \end{aligned}$$



Semantics informally:

- a conjunctive query

$$q(\mathbf{x}, s) \leftarrow \exists \mathbf{y}. \text{conj}(\mathbf{x}, \mathbf{y}) \wedge s = f(p_1(\mathbf{z}_1), \dots, p_n(\mathbf{z}_n))$$

is interpreted in an interpretation $\mathcal{I}$ as the set

$$q^{\mathcal{I}} = \{ \langle \mathbf{c}, v \rangle \in \Delta \times \dots \times \Delta \times [0, 1] \mid \dots$$

such that when we consider the substitution

$$\theta = \{ \mathbf{x}/\mathbf{c}, s/v \}$$

the formula

$$\exists \mathbf{y}. \text{conj}(\mathbf{x}, \mathbf{y}) \wedge s = f(p_1(\mathbf{z}_1), \dots, p_n(\mathbf{z}_n))$$

evaluates to true in $\mathcal{I}$.

- **Model of a query:** $\mathcal{I} \models q(\mathbf{c}, v)$ iff $\langle \mathbf{c}, v \rangle \in q^{\mathcal{I}}$
- **Entailment:** $\mathcal{K} \models q(\mathbf{c}, v)$ iff $\mathcal{I} \models \mathcal{K}$ implies $\mathcal{I} \models q(\mathbf{c}, v)$
- **Top-k retrieval:** $\text{ans}_{\text{top}-k}(\mathcal{K}, q) = \text{Top}_k\{ \langle \mathbf{c}, v \rangle \mid \mathcal{K} \models q(\mathbf{c}, v) \}$

How to determine the top- k answers of a query?

- Overall strategy: three steps

- 1 Check if $\mathcal{K}$ is satisfiable, as querying a non-satisfiable KB is meaningless (checkable in linear time)
- 2 Query q is *reformulated* into a set of conjunctive queries $r(q, \mathcal{T})$
 - ★ Basic idea: **reformulation procedure** closely resembles a top-down resolution procedure for logic programming

$$q(x, s) \leftarrow B(x) \wedge A(x) \wedge s = f(x)$$

$$B_1 \sqsubseteq A$$

$$B_2 \sqsubseteq A$$

$$q(x, s) \leftarrow B(x) \wedge B_1(x) \wedge s = f(x)$$

$$q(x, s) \leftarrow B(x) \wedge B_2(x) \wedge s = f(x)$$

- 3 The reformulated queries in $r(q, \mathcal{T})$ are evaluated over $\mathcal{A}$ (seen as a database) using standard top- k techniques for DBs
 - ★ for all $q_i \in r(q, \mathcal{T})$, $ans_{top-k}(q_i, \mathcal{A}) = \text{top-}k \text{ SQL query over } \mathcal{A} \text{ database}$
 - ★ $ans_{top-k}(KB, q) = Top_k(\bigcup_{q_i \in r(q, \mathcal{T})} ans_k(q_i, \mathcal{A}))$



Small Example:

P_2		B
0	s	1
3	t	2
4	q	5
6	q	7

$$\mathcal{T} = \{\exists P_2^- \sqsubseteq A, A \sqsubseteq \exists P_1, B \sqsubseteq \exists P_2\}$$

$$q(x, s) \leftarrow P_2(x, y) \wedge P_1(y, z) \wedge s = \max(0, 1 - x/10)$$

Small Example:

P_2		B
0	s	1
3	t	2
4	q	5
6	q	7

$$\mathcal{T} = \{\exists P_2^- \sqsubseteq A, A \sqsubseteq \exists P_1, B \sqsubseteq \exists P_2\}$$

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P_2		B
0	s	1
3	t	2
4	q	5
6	q	7

$$\mathcal{T} = \{\exists P_2^- \sqsubseteq A, A \sqsubseteq \exists P_1, B \sqsubseteq \exists P_2\}$$

$$q(x, s) \leftarrow P_2(x, y) \wedge P_1(y, z) \wedge s = \max(0, 1 - x/10)$$

$$q(x, s) \leftarrow P_2(x, y) \wedge A(y) \wedge s = \max(0, 1 - x/10)$$

Small Example:

P_2		B
0	s	1
3	t	2
4	q	5
6	q	7

$$\mathcal{T} = \{\exists \textcolor{red}{P}_2^- \sqsubseteq \textcolor{red}{A}, A \sqsubseteq \exists P_1, B \sqsubseteq \exists P_2\}$$

$$q(x, s) \leftarrow P_2(x, y) \wedge P_1(y, z) \wedge s = \max(0, 1 - x/10)$$

$$q(x, s) \leftarrow P_2(x, y) \wedge \textcolor{red}{A}(y) \wedge s = \max(0, 1 - x/10)$$

$$q(x, s) \leftarrow P_2(x, y) \wedge \textcolor{red}{P}_2(z, y) \wedge s = \max(0, 1 - x/10)$$

Small Example:

P_2		B
0	s	1
3	t	2
4	q	5
6	q	7

$$\mathcal{T} = \{\exists P_2^- \sqsubseteq A, A \sqsubseteq \exists P_1, B \sqsubseteq \exists P_2\}$$

$$q(x, s) \leftarrow P_2(x, y) \wedge P_1(y, z) \wedge s = \max(0, 1 - x/10)$$

$$q(x, s) \leftarrow P_2(x, y) \wedge A(y) \wedge s = \max(0, 1 - x/10)$$

$$q(x, s) \leftarrow P_2(x, y) \wedge P_2(z, y) \wedge s = \max(0, 1 - x/10)$$

$$q(x, s) \leftarrow P_2(x, y) \wedge s = \max(0, 1 - x/10)$$

Small Example:

P_2		B
0	s	1
3	t	2
4	q	5
6	q	7

$$\mathcal{T} = \{\exists P_2^- \sqsubseteq A, A \sqsubseteq \exists P_1, \textcolor{red}{B} \sqsubseteq \exists \textcolor{red}{P}_2\}$$

$$q(x, s) \leftarrow P_2(x, y) \wedge P_1(y, z) \wedge s = \max(0, 1 - x/10)$$

$$q(x, s) \leftarrow P_2(x, y) \wedge A(y) \wedge s = \max(0, 1 - x/10)$$

$$q(x, s) \leftarrow P_2(x, y) \wedge P_2(z, y) \wedge s = \max(0, 1 - x/10)$$

$$q(x, s) \leftarrow \textcolor{red}{P}_2(x, y) \wedge s = \max(0, 1 - x/10)$$

$$q(x, s) \leftarrow \textcolor{red}{B}(x) \wedge s = \max(0, 1 - x/10)$$

Small Example:

P_2		B
0	s	1
3	t	2
4	q	5
6	q	7

$$\mathcal{T} = \{\exists P_2^- \sqsubseteq A, A \sqsubseteq \exists P_1, B \sqsubseteq \exists P_2\}$$

$$q(x, s) \leftarrow P_2(x, y) \wedge P_1(y, z) \wedge s = \max(0, 1 - x/10)$$

$$q(x, s) \leftarrow P_2(x, y) \wedge A(y) \wedge s = \max(0, 1 - x/10)$$

$$q(x, s) \leftarrow P_2(x, y) \wedge P_2(z, y) \wedge s = \max(0, 1 - x/10)$$

$$q(x, s) \leftarrow P_2(x, y) \wedge s = \max(0, 1 - x/10)$$

$$q(x, s) \leftarrow B(x) \wedge s = \max(0, 1 - x/10)$$

$$q_1(x, s) \leftarrow P_2(x, y) \wedge s = \max(0, 1 - x/10)$$

$$q_2(x, s) \leftarrow B(x) \wedge s = \max(0, 1 - x/10)$$

Small Example:

P_2		B
0	s	1
3	t	2
4	q	5
6	q	7

$$\mathcal{T} = \{\exists P_2^- \sqsubseteq A, A \sqsubseteq \exists P_1, B \sqsubseteq \exists P_2\}$$

$$q(x, s) \leftarrow P_2(x, y) \wedge P_1(y, z) \wedge s = \max(0, 1 - x/10)$$

$$q(x, s) \leftarrow P_2(x, y) \wedge A(y) \wedge s = \max(0, 1 - x/10)$$

$$q(x, s) \leftarrow P_2(x, y) \wedge P_2(z, y) \wedge s = \max(0, 1 - x/10)$$

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$$q(x, s) \leftarrow B(x) \wedge s = \max(0, 1 - x/10)$$

$$q_1(x, s) \leftarrow P_2(x, y) \wedge s = \max(0, 1 - x/10)$$

$$q_2(x, s) \leftarrow B(x) \wedge s = \max(0, 1 - x/10)$$

$$ans_{top-3}(\mathcal{A}, q_1) = [\langle 0, 1.0 \rangle, \langle 3, 0.7 \rangle, \langle 4, 0.6 \rangle]$$

$$ans_{top-3}(\mathcal{A}, q_2) = [\langle 1, 0.9 \rangle, \langle 2, 0.8 \rangle, \langle 5, 0.5 \rangle]$$

Small Example:

P_2		B
0	s	1
3	t	2
4	q	5
6	q	7

$$\mathcal{T} = \{\exists P_2^- \sqsubseteq A, A \sqsubseteq \exists P_1, B \sqsubseteq \exists P_2\}$$

$$q(x, s) \leftarrow P_2(x, y) \wedge P_1(y, z) \wedge s = \max(0, 1 - x/10)$$

$$q(x, s) \leftarrow P_2(x, y) \wedge A(y) \wedge s = \max(0, 1 - x/10)$$

$$q(x, s) \leftarrow P_2(x, y) \wedge P_2(z, y) \wedge s = \max(0, 1 - x/10)$$

$$q(x, s) \leftarrow P_2(x, y) \wedge s = \max(0, 1 - x/10)$$

$$q(x, s) \leftarrow B(x) \wedge s = \max(0, 1 - x/10)$$

$$q_1(x, s) \leftarrow P_2(x, y) \wedge s = \max(0, 1 - x/10)$$

$$q_2(x, s) \leftarrow B(x) \wedge s = \max(0, 1 - x/10)$$

$$ans_{top-3}(\mathcal{A}, q_1) = [\langle 0, 1.0 \rangle, \langle 3, 0.7 \rangle, \langle 4, 0.6 \rangle]$$

$$ans_{top-3}(\mathcal{A}, q_2) = [\langle 1, 0.9 \rangle, \langle 2, 0.8 \rangle, \langle 5, 0.5 \rangle]$$

$$ans_{top-k}(\mathcal{K}, q) = [\langle 0, 1.0 \rangle, \langle 1, 0.9 \rangle, \langle 2, 0.8 \rangle]$$

Proposition

Given a DL-Lite KB $\mathcal{K} = \langle \mathcal{T}, \mathcal{A} \rangle$ and a query q then we can compute $ans_{top-k}(\mathcal{K}, q)$ in (sub) linear time w.r.t. the size of $\mathcal{A}$. The same holds for the description logic DLR-Lite.

Propositional Fuzzy Logics Basics

- **Formulae**: propositional formulae
- **Truth space** is $[0, 1]$
- **Formulae** have a degree of truth in $[0, 1]$
- **Interpretation**: is a mapping $\mathcal{I} : Atoms \rightarrow [0, 1]$
- Interpretations are **extended** to formulae using **norms** to interpret connectives

negation

$$n(0) = 1$$

$$a \leq b \text{ implies } n(b) \leq n(a)$$

s-norm (disjunction)

$$s(a, 0) = a$$

$$b \leq c \text{ implies } s(a, b) \leq s(a, c)$$

$$s(a, b) = s(b, a)$$

$$s(a, s(b, c)) = s(s(a, b), c)$$

t-norm (conjunction)

$$t(a, 1) = a$$

$$b \leq c \text{ implies } t(a, b) \leq t(a, c)$$

$$t(a, b) = t(b, a)$$

$$t(a, t(b, c)) = t(t(a, b), c)$$

i-norm (implication)

$$a \leq b \text{ implies } i(a, c) \geq i(b, c)$$

$$b \leq c \text{ implies } i(a, b) \leq i(a, c)$$

$$i(0, b) = 1$$

$$i(a, 1) = 1$$

Usually,

$$i(a, b) = \sup\{c : t(a, c) \leq b\}$$

$i(a, b) = \sup\{c : t(a, c) \leq b\}$ is called **r-implication** and depends on the t-norm only

Typical norms

	Lukasiewicz Logic	Gödel Logic	Product Logic	Zadeh
$\neg x$	$1 - x$	if $x = 0$ then 1 else 0	if $x = 0$ then 1 else 0	$1 - x$
$x \wedge y$	$\max(x + y - 1, 0)$	$\min(x, y)$	$x \cdot y$	$\min(x, y)$
$x \vee y$	$\min(x + y, 1)$	$\max(x, y)$	$x + y - x \cdot y$	$\max(x, y)$
$x \Rightarrow y$	if $x \leq y$ then 1 else $1 - x + y$	if $x \leq y$ then 1 else y	if $x \leq y$ then 1 else y/x	$\max(1 - x, y)$

Note: for Lukasiewicz Logic and Zadeh, $x \Rightarrow y \equiv \neg x \vee y$

$$\mathcal{I}(\phi \wedge \psi) = \mathcal{I}(\phi) \wedge \mathcal{I}(\psi)$$

$$\mathcal{I}(\phi \vee \psi) = \mathcal{I}(\phi) \vee \mathcal{I}(\psi)$$

$$\mathcal{I}(\phi \rightarrow \psi) = \mathcal{I}(\phi) \rightarrow \mathcal{I}(\psi)$$

$$\mathcal{I} \models \phi \quad \text{iff} \quad \mathcal{I}(\phi) = 1 \quad \text{iff} \quad \phi \text{ satisfiable}$$

$$\mathcal{I} \models \mathcal{T} \quad \text{iff} \quad \mathcal{I} \models \phi \text{ for all } \phi \in \mathcal{T}$$

$$\models \phi \quad \text{iff} \quad \text{for all } \mathcal{I} . \mathcal{I} \models \phi$$

$$\mathcal{T} \models \phi \quad \text{iff} \quad \text{for all } \mathcal{I} . \text{ if } \mathcal{I} \models \mathcal{T} \text{ then } \mathcal{I} \models \phi$$



- Note:

$$\begin{aligned}\neg\phi & \text{ is } \phi \rightarrow 0 \\ \phi \bar{\wedge} \psi & \text{ is } \phi \wedge (\phi \rightarrow \psi) \\ \phi \bar{\vee} \psi & \text{ is } ((\phi \rightarrow \psi) \rightarrow \psi) \bar{\wedge} ((\psi \rightarrow \phi) \rightarrow \phi) \\ \mathcal{I}(\phi \bar{\wedge} \psi) & = \min(\mathcal{I}(\phi), \mathcal{I}(\psi)) \\ \mathcal{I}(\phi \bar{\vee} \psi) & = \max(\mathcal{I}(\phi), \mathcal{I}(\psi))\end{aligned}$$

- Zadeh semantics: not interesting for fuzzy logicians: its a sub-logic of Łukasiewicz

$$\begin{aligned}\neg_Z \phi & = \neg_{\mathbf{L}} \phi \\ \phi \wedge_Z \psi & = \phi \wedge_{\mathbf{L}} (\phi \rightarrow_{\mathbf{L}} \psi) \\ \phi \rightarrow \psi & = \neg_{\mathbf{L}} \phi \vee_{\mathbf{L}} \psi\end{aligned}$$

- Hence, rarely considered by fuzzy logicians

Axioms of logic BL (Basic Fuzzy Logic)

Fix arbitrary t-norm and r-implication.

$$(A1) \quad (\phi \rightarrow \psi) \rightarrow ((\psi \rightarrow \chi) \rightarrow \phi \rightarrow \chi)$$

$$(A2) \quad (\phi \wedge \psi) \rightarrow \phi$$

$$(A3) \quad (\phi \wedge \psi) \rightarrow (\psi \wedge \phi)$$

$$(A4) \quad (\phi \wedge (\phi \rightarrow \psi)) \rightarrow (\psi \wedge (\psi \rightarrow \phi))$$

$$(A5a) \quad (\phi \wedge (\psi \rightarrow \chi)) \rightarrow ((\psi \wedge \psi) \rightarrow \chi))$$

$$(A5b) \quad ((\psi \wedge \psi) \rightarrow \chi)) \rightarrow (\phi \wedge (\psi \rightarrow \chi))$$

$$(A6) \quad (\phi \wedge (\psi \rightarrow \chi)) \rightarrow (((\psi \rightarrow \phi) \rightarrow \chi)) \rightarrow \chi)$$

$$(A7) \quad 0 \rightarrow \phi$$

(Deduction rule) Modus ponens: from ϕ and $\phi \rightarrow \psi$ infer ψ

Proposition

$\mathcal{T} \vdash_{BL} \phi$ iff $\mathcal{T} \models_{BL} \phi$. Also, if $\mathcal{T} \vdash_{BL} \phi$ then $\mathcal{T} \models_{BL2} \phi$, but not vice-versa (e.g. $\models_{BL2} \phi \vee \neg\phi$, but $\not\models_{BL} \phi \vee \neg\phi$).

- $\models_{BL} \phi \wedge \neg\phi \rightarrow 0$
- $\models_{BL} \phi \rightarrow \neg\neg\phi$, but $\not\models_{BL} \neg\neg\phi \rightarrow \phi$, e.g. $\phi = p \vee \neg p$, t-norm is Gödel
- $\models_{BL} (\phi \rightarrow \psi) \rightarrow (\neg\psi \rightarrow \neg\phi)$, but not vice-versa



Axioms of Łukasiewicz logic Ł

Fix Łukasiewicz t-norm and r-implication.

(Axioms) Axioms of BL

$$(Ł) \quad \neg\neg\phi \rightarrow \phi$$

(Deduction rule) Modus ponens: from ϕ and $\phi \rightarrow \psi$ infer ψ

Proposition

$\mathcal{T} \vdash_{Ł} \phi$ iff $\mathcal{T} \models_{Ł} \phi$.

- $\models_{Ł} \phi \rightarrow \psi \equiv \neg\psi \rightarrow \neg\phi$
- $\models_{Ł} \neg(\phi \wedge \psi) \equiv \neg\phi \vee \neg\psi$
- $\models_{Ł} \phi \rightarrow \psi \equiv \neg(\phi \wedge \neg\psi)$
- $\models_{Ł} \phi \rightarrow \psi \equiv \neg\phi \vee \neg\psi$
- $\models_{Ł} \neg(\phi \rightarrow \psi) \equiv \phi \wedge \neg\psi$
- Recall that “Zadeh logic” is a sub-logic of Ł

Axioms of Product logic Π

Fix product t-norm and r-implication.

(Axioms) Axioms of BL

$$(\Pi 1) \quad \neg\neg\chi \rightarrow ((\phi \wedge \chi \rightarrow \psi \wedge \chi) \rightarrow (\phi \rightarrow \psi))$$

$$(\Pi 2) \quad (\phi \bar{\wedge} \neg\phi) \rightarrow 0$$

(Deduction rule) Modus ponens: from ϕ and $\phi \rightarrow \psi$ infer ψ

Proposition

$\mathcal{T} \vdash_{\Pi} \phi$ iff $\mathcal{T} \models_{\Pi} \phi$.

- $\models_{\Pi} \neg(\phi \wedge \psi) \rightarrow \neg(\phi \bar{\wedge} \psi)$
- $\models_{\Pi} (\phi \rightarrow \neg\phi) \rightarrow \neg\phi$
- $\models_{\Pi} \neg\phi \bar{\vee} \neg\neg\phi$

Axioms of Gödel logic G

Fix Gödel t-norm and r-implication.

(Axioms) Axioms of BL

$$(G) \quad \phi \rightarrow (\phi \wedge \phi)$$

(Deduction rule) Modus ponens: from ϕ and $\phi \rightarrow \psi$ infer ψ

Proposition

$\mathcal{T} \vdash_G \phi$ iff $\mathcal{T} \models_G \phi$.

- $\models_G (\phi \wedge \psi) \equiv (\phi \bar{\wedge} \psi)$
- Gödel logic proves all axioms of intuitionistic logic I, vice-versa I + (A6) proves all axioms of Gödel logic

Axioms of Boolean logic

Fix interpretations to be boolean.

(Axioms) Axioms of BL

$$(BL2) \quad \phi \bar{\vee} \neg \phi$$

(Deduction rule) Modus ponens: from ϕ and $\phi \rightarrow \psi$ infer ψ

Proposition

$\mathcal{T} \vdash_{BL2} \phi$ iff $\mathcal{T} \models_{BL2} \phi$.

- $\models_{BL2} \phi \rightarrow (\phi \wedge \phi)$ (BL2 extends G)
- $\mathbb{L} + G$ is equivalent to BL2
- $\mathbb{L} + \Pi$ is equivalent to BL2
- $G + \Pi$ is equivalent to BL2

Axioms of Rational Pavelka Logic (RPL)

- Fix Łukasiewicz t-norm and r-implication
- Rational $r \in [0, 1]$ may appear as atom in formula. $\mathcal{I}(r) = r$
- Note: $\mathcal{I}(r \rightarrow \phi) = 1$ iff $\mathcal{I}(\phi) \geq r$. Also, $\mathcal{I}(\phi \rightarrow r) = 1$ iff $\mathcal{I}(\phi) \leq r$

(Axioms) Axioms of Ł

(Deduction rule) Modus ponens: from ϕ and $\phi \rightarrow \psi$ infer ψ

Proposition

$\mathcal{T} \vdash_{RPL} \phi$ iff $\mathcal{T} \models_{RPL} \phi$.

- RPL proves the derived deduction rule:
from $r \rightarrow \phi$ and $s \rightarrow (\phi \rightarrow \psi)$ infer $(r \wedge s) \rightarrow \psi$
- Let

$$||\phi||_{\mathcal{T}} = \inf\{\mathcal{I}(\phi) \mid \mathcal{I} \models \mathcal{T}\} \text{ (truth degree)}$$

$$|\phi|_{\mathcal{T}} = \sup\{r \mid \mathcal{T} \vdash r \rightarrow \phi\} \text{ (provability degree)}$$

then $||\phi||_{\mathcal{T}} = |\phi|_{\mathcal{T}}$

- Also,

$$|\neg\phi|_{\mathcal{T}} = 1 - |\phi|_{\mathcal{T}}$$

$$|\phi|_{\mathcal{T}} = \sup\{r \mid \mathcal{T} \vdash r \rightarrow \phi\} = \inf\{s \mid \mathcal{T} \vdash \phi \rightarrow s\}$$

Tableaux for Rational Pavelka Logic using MILP

Proposition

$|\phi|_{\mathcal{T}} = \min x$. such that $\mathcal{T} \cup \{\phi \rightarrow x\}$ satisfiable.

- We use MILP (Mixed Integer Linear Programming) to compute $|\phi|_{\mathcal{T}}$
- Let l be rational, variable or expression $1 - l'$ (l' variable), both admitting solution in $[0, 1]$, $\neg l = 1 - l$, $\neg \neg l = l$

$l \rightarrow p$	$\mapsto$	$x_p \geq l, x_p \in [0, 1]$
$p \rightarrow l$	$\mapsto$	$x_p \leq l, x_p \in [0, 1]$
$l \rightarrow \neg \phi$	$\mapsto$	$\phi \rightarrow \neg l$
$\neg \phi \rightarrow l$	$\mapsto$	$\neg l \rightarrow \phi$
$l \rightarrow (\phi \wedge \psi)$	$\mapsto$	$x_1 \rightarrow \phi, x_2 \rightarrow \psi, y \leq 1 - l, x_i \leq 1 - y, x_1 + x_2 = l + 1 - y, x_i \in [0, 1], y \in \{0, 1\}$
$(\phi \wedge \psi) \rightarrow l$	$\mapsto$	$x_1 \rightarrow \neg \phi, x_2 \rightarrow \neg \psi, x_1 + x_2 = 1 - l, x_i \in [0, 1]$
$l \rightarrow (\phi \rightarrow \psi)$	$\mapsto$	$\phi \rightarrow x_1, x_2 \rightarrow \psi, l + x_1 - x_2 = 1, x_i \in [0, 1]$
$(\phi \rightarrow \psi) \rightarrow l$	$\mapsto$	$x_1 \rightarrow \phi, \psi \rightarrow x_2, y - l \leq 0, y + x_1 \leq 1, y \leq x_2, y + l + x_1 - x_2 = 1, x_i \in [0, 1], y \in \{0, 1\}$

- After applying all the rules to $\mathcal{T} \cup \{\phi \rightarrow x\}$ (x variable), we have to solve a MILP problem of the form

$$\min \mathbf{c} \cdot \mathbf{x} \text{ s.t. } \mathbf{Ax} + \mathbf{By} \geq \mathbf{h}$$

where $a_{ij}, b_{ij}, c_i, h_k \in [0, 1]$, x_i admits solutions in $[0, 1]$, while y_j admits solutions in $\{0, 1\}$

Example

- Consider $\mathcal{T} = \{0.6 \rightarrow p, 0.7 \rightarrow (p \rightarrow q)\}$
- Let us show that $|q|_{\mathcal{T}} = 0.6 \wedge 0.7 = \max(1, 0.6 + 0.7 - 1) = 0.3$
- Recall that $|q|_{\mathcal{T}} = \min x.$ such that $\mathcal{T} \cup \{q \rightarrow x\}$

$$\mathcal{T} \cup \{q \rightarrow x\} = \{0.6 \rightarrow p, 0.7 \rightarrow (p \rightarrow q), q \rightarrow x, x \in [0, 1]\}$$

$$\mapsto \{x_p \geq 0.6, x_q \leq x, 0.7 \rightarrow (p \rightarrow q), \{x, x_p\} \subseteq [0, 1]\}$$

$$\mapsto \{x_p \geq 0.6, x_q \leq x, p \rightarrow x_1, x_2 \rightarrow q, 0.7 + x_1 - x_2 = 1, \{x, x_p, x_i\} \subseteq [0, 1]\}$$

$$\mapsto \{x_p \geq 0.6, x_q \leq x, x_p \leq x_1, x_q \geq x_2, 0.7 + x_1 - x_2 = 1, \{x, x_p, x_i\} \subseteq [0, 1]\} = S$$

It follows that $0.3 = \min x.$ such that $Sat(S)$

- Note:** A similar technique can be used for logic G and Π , but mixed integer non-linear programming is needed in place of MILP

Predicate Fuzzy Logics Basics

- **Formulae**: First-Order Logic formulae, *terms* are either variables or constants
 - ▶ we may introduce functions symbols as well, with crisp semantics (but uninteresting), or we need to discuss also fuzzy equality (which we leave out here)
- **Truth space** is $[0, 1]$
- **Formulae** have a degree of truth in $[0, 1]$
- **Interpretation**: is a mapping $\mathcal{I} : \text{Atoms} \rightarrow [0, 1]$
- Interpretations are **extended** to formulae as follows:

$$\begin{aligned}(\neg\phi &= \phi \rightarrow 0) \\ \mathcal{I}(\phi \wedge \psi) &= \mathcal{I}(\phi) \wedge \mathcal{I}(\psi) \\ \mathcal{I}(\phi \rightarrow \psi) &= \mathcal{I}(\phi) \rightarrow \mathcal{I}(\psi) \\ \mathcal{I}(\exists x\phi) &= \sup_{c \in \Delta^{\mathcal{I}}} \mathcal{I}_x^c(\phi) \\ \mathcal{I}(\forall x\phi) &= \inf_{c \in \Delta^{\mathcal{I}}} \mathcal{I}_x^c(\phi)\end{aligned}$$

where $\mathcal{I}_x^c$ is as $\mathcal{I}$, except that variable x is mapped into individual c

- Definitions of $\mathcal{I} \models \phi$, $\mathcal{I} \models \mathcal{T}$, $\models \phi$, $\mathcal{T} \models \phi$, $\|\phi\|_{\mathcal{I}}$ and $|\phi|_{\mathcal{T}}$ are as for the propositional case

Axioms of logic $\mathcal{C}\forall$, where $\mathcal{C} \in \{\text{BL}, \text{L}, \Pi, \text{G}\}$

(Axioms) Axioms of $\mathcal{C}$

($\forall 1$) $\forall x\phi(x) \rightarrow \phi(t)$ (t substitutable for x in $\phi(x)$)

($\exists 1$) $\phi(t) \rightarrow \exists x\phi(x)$ (t substitutable for x in $\phi(x)$)

($\forall 2$) $\forall x(\psi \rightarrow \phi) \rightarrow (\psi \rightarrow \forall x\phi)$ (x not free in ψ)

($\exists 2$) $\forall x(\phi \rightarrow \psi) \rightarrow (\exists x\phi \rightarrow \psi)$ (x not free in ψ)

($\forall 3$) $\forall x(\phi \bar{\vee} \psi) \rightarrow (\forall x\phi) \bar{\vee} \psi$ (x not free in ψ)

(Modus ponens) from ϕ and $\phi \rightarrow \psi$ infer ψ

(Generalization) from ϕ infer $\forall x\phi$

Proposition

$\mathcal{T} \vdash_{\mathcal{C}} \phi$ iff $\mathcal{T} \models_{\mathcal{C}} \phi$.

- if $\rightarrow$ is an r-implication then $\|\psi\|_{\mathcal{T}} \geq \|\phi\|_{\mathcal{T}} \wedge \|\phi \rightarrow \psi\|_{\mathcal{T}}$
- $\models_{\text{BL}\forall} \exists x\phi \rightarrow \neg\forall x\neg\phi$
- $\models_{\text{BL}\forall} \neg\exists x\phi \equiv \forall x\neg\phi$
- $\models_{\text{L}\forall} \exists x\phi \equiv \neg\forall x\neg\phi$

- $(\neg\forall x p(x)) \wedge (\neg\exists x \neg p(x))$ has no classical model. In Gödel logic it has no finite model, but has an **infinite** model: for integer $n \geq 1$, let $\mathcal{I}$ such that $p^{\mathcal{I}}(n) = 1/n$

$$\begin{aligned}(\forall x p(x))^{\mathcal{I}} &= \inf_n 1/n = 0 \\ (\exists x \neg p(x))^{\mathcal{I}} &= \sup_n \neg 1/n = \sup 0 = 0\end{aligned}$$

- **Note:** If $\mathcal{I} \models \exists x \phi(x)$ then not necessarily there is $c \in \Delta^{\mathcal{I}}$ such that $\mathcal{I} \models \phi(c)$.

$$\begin{aligned}\Delta^{\mathcal{I}} &= \{n \mid \text{integer } n \geq 1\} \\ p^{\mathcal{I}}(n) &= 1 - 1/n < 1, \text{ for all } n \\ (\exists x p(x))^{\mathcal{I}} &= \sup_n 1 - 1/n = 1\end{aligned}$$

- **Witnessed formula:** $\exists x \phi(x)$ is witnessed in $\mathcal{I}$ iff there is $c \in \Delta^{\mathcal{I}}$ such that $(\exists x \phi(x))^{\mathcal{I}} = (\phi(c))^{\mathcal{I}}$ (similarly for $\forall x \phi(x)$)
- **Witnessed interpretation:** $\mathcal{I}$ witnessed if all quantified formulae are witnessed in $\mathcal{I}$

Proposition

In $\mathcal{L}$, ϕ is satisfiable iff there is a witnessed model of ϕ .

The proposition does not hold for logic G and Π

Predicate Rational Pavelka Logic (RPL $\forall$)

- Fix Łukasiewicz t-norm and r-implication
- Formulae are as for Ł $\forall$, where rationals $r \in [0, 1]$ may appear as atoms

(Axioms and rules) As for Ł $\forall$

Proposition

$\mathcal{T} \vdash_{RPL\forall} \phi$ iff $\mathcal{T} \models_{RPL\forall} \phi$.

Fuzzy DLs Basics

- In classical DLs, a concept C is interpreted by an interpretation $\mathcal{I}$ as a set of individuals
- In fuzzy DLs, a concept C is interpreted by $\mathcal{I}$ as a fuzzy set of individuals
- Each individual is instance of a concept to a degree in $[0, 1]$
- Each pair of individuals is instance of a role to a degree in $[0, 1]$

Fuzzy $\mathcal{ALC}$

The semantics is an immediate consequence of the First-Order-Logic translation of DLs expressions

Interpretation:	$\mathcal{I}$	=	$\Delta^{\mathcal{I}}$	$\wedge$	=	t-norm
	$C^{\mathcal{I}}$	:	$\Delta^{\mathcal{I}} \rightarrow [0, 1]$	$\vee$	=	s-norm
	$R^{\mathcal{I}}$	:	$\Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}} \rightarrow [0, 1]$	$\neg$	=	negation
				$\rightarrow$	=	implication

Syntax		Semantics	
Concepts:	$C, D \rightarrow \top$	$\top^{\mathcal{I}}(x)$	= 1
	$\perp$	$\perp^{\mathcal{I}}(x)$	= 0
	A	$A^{\mathcal{I}}(x)$	$\in [0, 1]$
	$C \sqcap D$	$(C_1 \sqcap C_2)^{\mathcal{I}}(x)$	= $C_1^{\mathcal{I}}(x) \wedge C_2^{\mathcal{I}}(x)$
	$C \sqcup D$	$(C_1 \sqcup C_2)^{\mathcal{I}}(x)$	= $C_1^{\mathcal{I}}(x) \vee C_2^{\mathcal{I}}(x)$
	$\neg C$	$(\neg C)^{\mathcal{I}}(x)$	= $\neg C^{\mathcal{I}}(x)$
	$\exists R.C$	$(\exists R.C)^{\mathcal{I}}(x)$	= $\sup_{y \in \Delta^{\mathcal{I}}} R^{\mathcal{I}}(x, y) \wedge C^{\mathcal{I}}(y)$
	$\forall R.C$	$(\forall R.C)^{\mathcal{I}}(u)$	= $\inf_{y \in \Delta^{\mathcal{I}}} R^{\mathcal{I}}(x, y) \rightarrow C^{\mathcal{I}}(y)$

Assertions: $\langle a:C, r \rangle, \mathcal{I} \models \langle a:C, r \rangle$ iff $C^{\mathcal{I}}(a^{\mathcal{I}}) \geq r$ (similarly for roles)

- individual a is instance of concept C at least to degree $r, r \in [0, 1] \cap \mathbb{Q}$

Inclusion axioms: $C \sqsubseteq D$,

- $\mathcal{I} \models C \sqsubseteq D$ iff $\forall x \in \Delta^{\mathcal{I}}. C^{\mathcal{I}}(x) \leq D^{\mathcal{I}}(x)$
- this is equivalent to, $\forall x \in \Delta^{\mathcal{I}}. (C^{\mathcal{I}}(x) \rightarrow D^{\mathcal{I}}(x)) = 1$, if $\rightarrow$ is an r-implication

Basic Inference Problems

Consistency: Check if knowledge is meaningful

- Is $\mathcal{K}$ consistent, i.e. satisfiable?

Subsumption: structure knowledge, compute taxonomy

- $\mathcal{K} \models C \sqsubseteq D$?

Equivalence: check if two fuzzy concepts are the same

- $\mathcal{K} \models C = D$?

Graded instantiation: Check if individual a instance of class C to degree at least r

- $\mathcal{K} \models \langle a:C, r \rangle$?

BTVB: Best Truth Value Bound problem

- $|a:C|_{\mathcal{K}} = \sup\{r \mid \mathcal{K} \models \langle a:C, r \rangle\}$?

Top-k retrieval: Retrieve the top-k individuals that instantiate C w.r.t. best truth value bound

- $ans_{top-k}(\mathcal{K}, C) = Top_k\{\langle a, v \rangle \mid v = |a:C|_{\mathcal{K}}\}$

Some Notes on ...

- Value restrictions:
 - ▶ In classical DLs, $\forall R.C \equiv \neg \exists R. \neg C$
 - ▶ The same is not true, in general, in fuzzy DLs (depends on the operators' semantics, true for Łukasiewicz, but not true in Gödel logic)
 - ▶ Is it acceptable that $\forall hasParent.Human \not\equiv \neg \exists hasParent. \neg Human$?
Recall that in Ł and Zadeh, $\forall x. \phi \equiv \neg \exists x. \neg \phi$
- Models:
 - ▶ In classical DLs $\top \sqsubseteq \neg(\forall R.A) \sqcap (\neg \exists R. \neg A)$ has no classical model
 - ▶ In Gödel logic it has no finite model, but has an **infinite** model
- The **choice** of the appropriate semantics of the logical connectives is **important**.
 - ▶ Should have reasonable logical properties
 - ▶ **Certainly it must have efficient algorithms solving basic inference problems**
- **Łukasiewicz Logic** seems the best compromise, though Zadeh semantics has been considered historically in DLs (we recall that Zadeh semantics is not considered by fuzzy logicians)
- For disjointness it is better to use $C \sqcap D \sqsubseteq \perp$ rather than $C \sqsubseteq \neg D$
 - ▶ they are not the same, e.g. $A \sqsubseteq \neg A$ says that $A^{\mathcal{I}}(x) \leq 0.5$ holds, for all $\mathcal{I}$ and for all $x \in \Delta^{\mathcal{I}}$ (under Łukasiewicz Logic)

Towards fuzzy OWL Lite and OWL DL

- Recall that OWL Lite and OWL DL relate to $SHIF(D)$ and $SHOIN(D)$, respectively
- We need to extend the semantics of fuzzy ALC to fuzzy $SHOIN(D) = ALCHOIN\mathcal{R}_+(D)$
- Additionally, we add
 - ▶ **modifiers** (e.g., *very*)
 - ▶ **concrete fuzzy concepts** (e.g., *Young*)
 - ▶ both additions have **explicit** membership functions

Number Restrictions, Inverse and Transitive roles

- The semantics of the concept $(\geq n R)$ is:

$$(\geq n R)^{\mathcal{I}}(x) = \sup_{\{y_1, \dots, y_n\} \subseteq \Delta^{\mathcal{I}}} \bigwedge_{i=1}^n R^{\mathcal{I}}(x, y_i)$$

- It is the result of viewing $(\geq n R)$ as the open first order formula

$$\exists y_1, \dots, y_n. \bigwedge_{i=1}^n R(x, y_i) \wedge \bigwedge_{1 \leq i < j \leq n} y_i \neq y_j.$$

- The semantics of the concept $(\leq n R)$ is:

$$(\leq n R)^{\mathcal{I}}(x) = \neg(\geq n+1 R)^{\mathcal{I}}(x)$$

- Note: $(\geq 1 R) \equiv \exists R.$

- For transitive roles we have for all $x, y \in \Delta^{\mathcal{I}}$

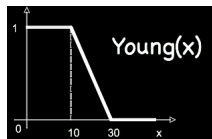
$$R^{\mathcal{I}}(x, y) = R^{\mathcal{I}}(y, x)$$

- For transitive roles R we impose: for all $x, y \in \Delta^{\mathcal{I}}$

$$R^{\mathcal{I}}(x, y) \geq \sup_{z \in \Delta^{\mathcal{I}}} \min(R^{\mathcal{I}}(x, z), R^{\mathcal{I}}(z, y))$$

Concrete fuzzy concepts

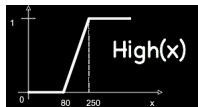
- E.g., *Small*, *Young*, *High*, etc. with **explicit** membership function
- Use the idea of concrete domains:
 - ▶ $D = \langle \Delta_D, \Phi_D \rangle$
 - ▶ Δ_D is an interpretation domain
 - ▶ Φ_D is the set of concrete fuzzy domain predicates d with a predefined arity $n = 1, 2$ and **fixed** interpretation $d^D: \Delta_D^n \rightarrow [0, 1]$
 - ▶ For instance,



$$\begin{aligned} \text{Minor} &= \text{Person} \sqcap \exists \text{hasAge.} \leq_{18} \\ \text{YoungPerson} &= \text{Person} \sqcap \exists \text{hasAge.} \text{Young} \\ &\quad \text{functional}(\text{hasAge}) \end{aligned}$$

Modifiers

- *Very*, *moreOrLess*, *slightly*, etc.
- Apply to fuzzy sets to change their membership function
 - ▶ $very(x) = x^2$
 - ▶ $slightly(x) = \sqrt{x}$
- For instance,



$$SportsCar = Car \cap \exists speed.very(High)$$

Fuzzy *SHOIN*(D)

Concepts:

	Syntax	Semantics
$C, D \longrightarrow$	$\top$	$\top(x)$
	$\perp$	$\perp(x)$
	A	$A(x)$
	$(C \sqcap D)$	$C_1(x) \wedge C_2(x)$
	$(C \sqcup D)$	$C_1(x) \vee C_2(x)$
	$(\neg C)$	$\neg C(x)$
	$(\exists R.C)$	$\exists x R(x, y) \wedge C(y)$
	$(\forall R.C)$	$\forall x R(x, y) \rightarrow C(y)$
	$\{a\}$	$x = a$
	$(\geq n R)$	$\exists y_1, \dots, y_n. \bigwedge_{i=1}^n R(x, y_i) \wedge \bigwedge_{1 \leq i < j \leq n} y_i \neq y_j$
	$(\leq n R)$	$\neg(\geq n+1 R)(x)$
	FCC	$\mu_{FCC}(x)$
	$M(C)$	$\mu_M(C(x))$
$R \longrightarrow$	P	$P(x, y)$
	P^-	$P(y, x)$

Assertions:

	Syntax	Semantics
$\alpha \longrightarrow$	$\langle a:C, r \rangle$	$r \rightarrow C(a)$
	$\langle (a, b):R, r \rangle$	$r \rightarrow R(a, b)$

Axioms:

	Syntax	Semantics
$\tau \longrightarrow$	$C \sqsubseteq D$	$\forall x C(x) \rightarrow D(x) = 1$, where $\rightarrow$ is r-implication
	$fun(R)$	$\forall x \forall y \forall z R(x, y) \wedge R(x, z) \rightarrow y = z$
	$trans(R)$	$(\exists z R(x, z) \wedge R(z, y)) \rightarrow R(x, y)$

Reasoning

Depends on the semantics and reasoning method (tableau-based or MILP-based)

Tableaux method: under Zadeh semantics

- a tableau exists for fuzzy *SHIN*, solving the satisfiability problem
- classical blocking methods apply similarly in the fuzzy variant
- the management of General concept inclusions (GCI's) is more complicated compared to the crisp case
- a translation of fuzzy *SHOIN* to crisp *SHOIN* also exists (not addressed here)
- the tableaux method is **not suitable** to deal with fuzzy concrete concepts and modifiers
- the BTVB can be solved, but not efficiently

MILP based method: under Zadeh semantics, Łukasiewicz semantics, and classical semantics

- **exists** for fuzzy *ALC* + linear modifiers + fuzzy concrete concepts (published)
- **exists** for fuzzy *SHIF* + linear modifiers + fuzzy concrete concepts (implemented, but not published yet)
- solves the BTVB as primary problem

Fuzzy tableaux-based method

- **Tableau algorithm is similar to classical DL tableaux**
- Most problems can be reduced to satisfiability problem, e.g.
- Assertions are extended to $\langle a:C \geq n \rangle$, $\langle a:C \leq n \rangle$, $\langle a:C > n \rangle$ and $\langle a:C < n \rangle$
- $\mathcal{K} \models \langle a:C, n \rangle$ iff $\mathcal{K} \cup \{\langle a:C < n \rangle\}$ not satisfiable
 - ▶ All models of $\mathcal{K}$ do not satisfy $\langle a:C < n \rangle$, i.e. do satisfy $\langle a:C \geq n \rangle$
- Let's see a tableaux algorithm for satisfiability checking, where

$$x \wedge y = \min(x, y)$$

$$x \vee y = \max(x, y)$$

$$\neg x = 1 - x$$

$$x \rightarrow y = \max(1 - x, y)$$

Tableaux for $\mathcal{ALC}$ KB

- Works on a tree forest (semantics through viewing tree as an ABox)
 - ▶ Nodes represent elements of $\Delta^{\mathcal{I}}$, labelled with sub-concepts of C and their weights
 - ▶ Edges represent role-successorships between elements of $\Delta^{\mathcal{I}}$ and their weights
- Works on concepts in **negation normal form**: push negation inside using de Morgan's laws and

$$\neg(\exists R.C) \mapsto \forall R.\neg C$$

$$\neg(\forall R.C) \mapsto \exists R.\neg C$$

- It is initialised with a tree forest consisting of root nodes a , for all individuals appearing in the KB:
 - ▶ If $\langle a:C \bowtie n \rangle \in \mathcal{K}$ then $\langle C, \bowtie, n \rangle \in \mathcal{L}(a)$
 - ▶ If $\langle (a,b):R \bowtie n \rangle \in \mathcal{K}$ then $\langle \langle a,b \rangle, \bowtie, n \rangle \in \mathcal{E}(R)$
- A tree forest T contains a **clash** if for a tree T in the forest there is a node x in T , containing a **conjugated pair** $\{\langle A, \triangleright, n \rangle, \langle C, \triangleleft, m \rangle\} \subseteq \mathcal{L}(x)$, e.g. $\langle A, \geq, 0.6 \rangle, \langle A, <, 0.3 \rangle$
- Returns “ $\mathcal{K}$ is satisfiable” if rules can be applied s.t. they yield a clash-free, complete (no more rules apply) tree forest

$\mathcal{ALC}$ Tableau rules (excerpt)

$x \bullet \{\langle C_1 \sqcap C_2, \geq, n \rangle, \dots\}$	$\longrightarrow \sqcap$	$x \bullet \{\langle C_1 \sqcap C_2, \geq, n \rangle, \langle C_1, \geq, n \rangle, \langle C_2, \geq, n \rangle, \dots\}$
$x \bullet \{\langle C_1 \sqcup C_2, \geq, n \rangle, \dots\}$	$\longrightarrow \sqcup$	$x \bullet \{\langle C_1 \sqcup C_2, \geq, n \rangle, \langle C, \geq, n \rangle, \dots\}$ for $C \in \{C_1, C_2\}$
$x \bullet \{\langle \exists R.C, \geq, n \rangle, \dots\}$	$\longrightarrow \exists$	$x \bullet \{\langle \exists R.C, \geq, n \rangle, \dots\}$ $\langle R, \geq, n \rangle \downarrow$ $y \bullet \{\langle C, \geq, n \rangle\}$
$x \bullet \{\langle \forall R.C, \geq, n \rangle, \dots\}$ $\langle R, \geq, m \rangle \downarrow \quad (m > 1 - n)$ $y \bullet \{\dots\}$	$\longrightarrow \forall$	$x \bullet \{\langle \forall R.C, \geq, n \rangle, \dots\}$ $\langle R, \geq, m \rangle \downarrow$ $y \bullet \{\dots, \langle C, \geq, n \rangle\}$
$x \bullet \{C \sqsubseteq D, \dots\}$	$\longrightarrow \sqsubseteq$	$x \bullet \{C \sqsubseteq D, E, \dots\}$ for $E \in \{\langle C, <, n \rangle, \langle D, \geq, n \rangle\}, n \in N^A$
$\vdots$	$\vdots$	$\vdots$

$$\begin{aligned}
 \mathcal{K} &= \langle \mathcal{T}, \mathcal{A} \rangle \\
 X^{\mathcal{A}} &= \{0, 0.5, 1\} \cup \{n \mid \langle \alpha \bowtie n \rangle \in \mathcal{A}\} \\
 N^{\mathcal{A}} &= X^{\mathcal{A}} \cup \{1 - n \mid n \in X^{\mathcal{A}}\}
 \end{aligned}$$

Theorem

Let $\mathcal{K}$ be an $\mathcal{ALC}$ KB and F obtained by applying the tableau rules to $\mathcal{K}$. Then

- 1 The rule application terminates,
- 2 If F is clash-free and complete, then F defines a (canonical) (tree forest) model for $\mathcal{K}$, and
- 3 If $\mathcal{K}$ has a model $\mathcal{I}$, then the rules can be applied such that they yield a clash-free and complete forest F .

The tableau can be modified to a decision procedure for

- $SHIN (\equiv \mathcal{ALCHINR}_+)$

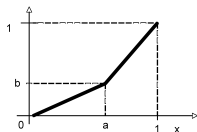
Problem with fuzzy tableau

- Usual fuzzy tableaux calculus **does not work anymore** with
 - ▶ modifiers and concrete fuzzy concepts
 - ▶ Łukasiewicz Logic
- Usual fuzzy tableaux calculus does not solve the BTVB problem
- New algorithm uses **bounded Mixed Integer Programming oracle**, as for Many Valued Logics
 - ▶ Recall: the *general MILP problem* is to find

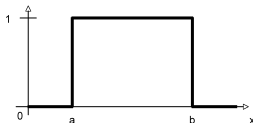
$$\begin{aligned}\bar{\mathbf{x}} &\in \mathbb{Q}^k, \bar{\mathbf{y}} \in \mathbb{Z}^m \\ f(\bar{\mathbf{x}}, \bar{\mathbf{y}}) &= \min\{f(\mathbf{x}, \mathbf{y}) : A\mathbf{x} + B\mathbf{y} \geq \mathbf{h}\} \\ A, B &\text{ integer matrixes}\end{aligned}$$

Requirements

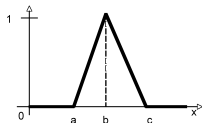
- Works for usual fuzzy DL semantics (Zadeh semantics) and Lukasiewicz logic
- Modifiers are definable as linear in-equations over $\mathbb{Q}, \mathbb{Z}$ (e.g., linear hedges), for instance, linear hedges, $lm(a, b)$, e.g. *very* = $lm(0.7, 0.49)$
- Fuzzy concrete concepts are definable as linear in-equations over $\mathbb{Q}, \mathbb{Z}$ (e.g., crisp, triangular, trapezoidal, left shoulder and right shoulder membership functions)



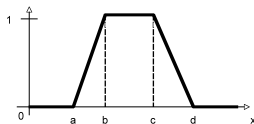
$lm(a, b)$



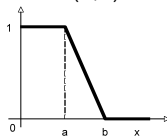
$cr(a, b)$



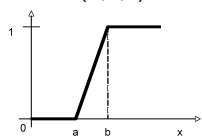
$tri(a, b, c)$



$trz(a, b, c, d)$

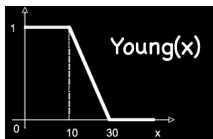


$ls(a, b)$



$rs(a, b, c)$

- Example:



$$\begin{aligned}
 \text{Minor} &= \text{Person} \sqcap \exists \text{hasAge.} \leq_{18} \\
 \text{YoungPerson} &= \text{Person} \sqcap \exists \text{hasAge. Young} \\
 \text{Young} &= \text{Is}(10, 30) \\
 \leq_{18} &= \text{cr}(0, 18)
 \end{aligned}$$

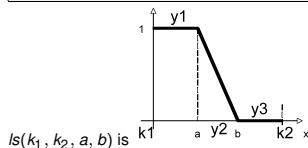
- Then

$$\begin{aligned}
 |a:C|_{\mathcal{K}} &= \min\{x \mid \mathcal{K} \cup \{\langle a:C \leq x \rangle \text{ satisfiable} \}\} \\
 |C \sqsubseteq D|_{\mathcal{K}} &= \min\{x \mid \mathcal{K} \cup \{\langle a:C \sqcap \neg D \geq 1 - x \rangle \text{ satisfiable} \}\}
 \end{aligned}$$

- Apply (**deterministic**) tableaux calculus, then use bounded Mixed Integer Programming oracle

$\mathcal{ALC}$ MILP Tableau rules under Zadeh semantics (excerpt)

$x \bullet \{\langle C_1 \sqcap C_2, \geq, l \rangle, \dots\}$	$\longrightarrow \sqcap$	$x \bullet \{\langle C_1 \sqcap C_2, \geq, l \rangle, \langle C_1, \geq, l \rangle, \langle C_2, \geq, l \rangle, \dots\}$
$x \bullet \{\langle C_1 \sqcup C_2, \geq, l \rangle, \dots\}$	$\longrightarrow \sqcup$	$x \bullet \{\langle C_1 \sqcup C_2, \geq, l \rangle, \langle C_1, \geq, x_1 \rangle, \langle C_2, \geq, x_2 \rangle, \\ x_1 + x_2 = l, x_1 \leq y, x_2 \leq 1 - y, \\ x_i \in [0, 1], y \in \{0, 1\}, \dots\}$
$x \bullet \{\langle \exists R.C, \geq, l \rangle, \dots\}$	$\longrightarrow \exists$	$x \bullet \{\langle \exists R.C, \geq, l \rangle, \dots\}$ $\langle R, \geq, l \rangle \downarrow$ $y \bullet \{\langle C, \geq, l \rangle\}$
$x \bullet \{\langle \forall R.C, \geq, l_1 \rangle, \dots\}$ $\langle R, \geq, l_2 \rangle \downarrow$ $y \bullet \{\dots\}$	$\longrightarrow \forall$	$x \bullet \{\langle \forall R.C, \geq, l_1 \rangle, \dots\}$ $\langle R, \geq, l_2 \rangle \downarrow$ $y \bullet \{\dots, \langle C, \geq, x \rangle$ $x + y \geq l_1, x \leq y, l_1 + l_2 \leq 2 - y,$ $x \in [0, 1], y \in \{0, 1\}\}$
$x \bullet \{A \sqsubseteq C, \langle A, \geq, l \rangle, \dots\}$	$\longrightarrow \sqsubseteq_1$	$x \bullet \{A \sqsubseteq C, \langle C, \geq, l \rangle, \dots\}$
$x \bullet \{C \sqsubseteq A, \langle A, \leq, l \rangle, \dots\}$	$\longrightarrow \sqsubseteq_2$	$x \bullet \{C \sqsubseteq A, \langle C, \leq, l \rangle, \dots\}$
$x \bullet \{C \sqsubseteq D, \dots\}$	$\longrightarrow \sqsubseteq$	$x \bullet \{C \sqsubseteq D, \langle C, \leq, x \rangle, \langle D, \geq, x \rangle, x \in [0, 1], \dots\}$
$x \bullet \{\langle ls(k_1, k_2, a, b), \geq, l \rangle, \dots\}$	$\longrightarrow \sqsubseteq$	$x \bullet \{ls(k_1, k_2, a, b), y_1 + y_2 + y_3 = 1, y_i \in \{0, 1\}, \\ x + (k_2 - a) \cdot y_1 \leq k_2, x + (k_1 - a) \cdot y_2 \geq k_1, \\ x + (k_2 - b) \cdot y_2 \geq k_2, \\ x + (b - a) \cdot l + (k_2 - a) \cdot y_2 \leq k_2 - a + b, \\ x + (k_1 - b) \cdot y_3 \leq k_1, l + y_3 \leq 1, \dots\}$
$\vdots$	$\vdots$	$\vdots$



Example

• Suppose $\mathcal{K} = \begin{cases} A \sqcap B \sqsubseteq C \\ \langle a:A \geq 0.3 \rangle \\ \langle a:B \geq 0.4 \rangle \end{cases}$

Query : = $|a:C|_{\mathcal{K}} = \min\{x \mid \mathcal{K} \cup \{\langle a:C \leq x \rangle\} \text{ satisfiable}\}$

Step	Tree	
1.	$a \bullet \{\langle A, \geq, 0.3 \rangle, \langle B, \geq, 0.4 \rangle, \langle C, \leq, x \rangle\}$	(Hypothesis)
2.	$\cup \{\langle A \sqcap B, \leq, x \rangle\}$	$(\rightarrow \sqsubseteq_2)$
3.	$\cup \{\langle A, \leq, x_1 \rangle, \langle B, \leq, x_2 \rangle\}$ $\cup \{x = x_1 + x_2 - 1, 1 - y \leq x_1, y \leq x_2\}$ $\cup \{x_i \in [0, 1], y \in \{0, 1\}\}$	$(\rightarrow \sqcap_{\leq})$
4.	find $\min\{x \mid \langle a:A \geq 0.3 \rangle, \langle a:B \geq 0.4 \rangle,$ $\langle a:C \leq x \rangle, \langle a:A \leq x_1 \rangle, \langle a:B \leq x_2 \rangle,$ $x = x_1 + x_2 - 1, 1 - y \leq x_1, y \leq x_2,$ $x_i \in [0, 1], y \in \{0, 1\}\}$	(MILP Oracle)
5.	MILP oracle: $\mathbf{x = 0.3}$	

Implementation issues

- Several options exists:
 - ▶ Try to map fuzzy DLs to classical DLs
 - ★ but, does not work with modifiers and concrete fuzzy concepts
 - ▶ Try to map fuzzy DLs to some fuzzy logic programming framework
 - ★ A lot of work exists about mappings among classical DLs and LPs
 - ★ But, needs a theorem prover for fuzzy LPs (not addressed here)
 - ★ To be used then e.g. in the axiomatic approach to fuzzy DLPs (Description Logic Programs)
 - ▶ Build an ad-hoc theorem prover for fuzzy DLs, using e.g., MILP
 - ★ To be used then separately e.g. in the DL-log approach to fuzzy DLPs
- A theorem prover for fuzzy *SHIF* + linear hedges + concrete fuzzy concepts, using MILP, has been implemented (<http://gaia.isti.cnr.it/~straccia>)

Future Work on fuzzy DLs

- Research directions:

- ▶ Computational complexity of the fuzzy DLs family
- ▶ Design of efficient reasoning algorithms
- ▶ Combining fuzzy DLs with fuzzy Logic Programming
- ▶ Language extensions: e.g. fuzzy quantifiers

TopCustomer = Customer $\sqcap$ (Usually)buys.Expensiveltem
Expensiveltem = Item $\sqcap$ $\exists$ price.High

- ▶ Conjunctive query answering (top-k query answering) for more expressive DLs
- ▶ Developing systems, extending **fuzzyDL system**, ...
- ▶ Applications, e.g. Ontology mediated data access ((distributed) multimedia information retrieval, resource selection, ...), Negotiation, ...
- ▶ ...